Subject: Mathematics | **WEEKLY CHAPTER TEST – 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

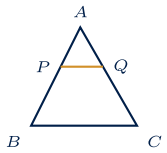
1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

1. A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, the length of her shadow after 4 seconds is:

(a) 1.2 m (b) 1.6 m (c) 2.0 m (d) 2.4 m

2. In ΔABC, P and Q are points on AB and AC respectively such that $PQ \parallel BC$. If $AP = 2.4$ cm, $AQ = 2.0$ cm, and $QC = 3.0$ cm, then the length of AB is:



(a) 4.8 cm (b) 6.0 cm (c) 5.0 cm (d) 7.2 cm

3. Which of the following is **NOT** a valid criterion for similarity of two triangles?

(a) SAS (b) AAA (c) SSA (d) SSS

4. If in triangles ABC and PQR, $\angle A = \angle Q = 40°$, $AB : QR = AC : QP$, and $\angle R = 65°$, then the degree measure of $\angle B$ is:

(a) 65° (b) 75° (c) 40° (d) 85°

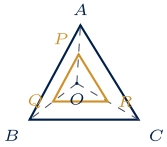
5. **Assertion (A):** If in ΔABC and ΔPQR, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, then $\Delta ABC \sim \Delta PQR$.
Reason (R): If corresponding sides of two triangles are proportional, then their corresponding angles are equal and the triangles are similar.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

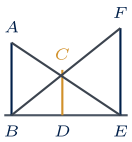
SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

6. **[2 Marks]** In ΔABC, $\angle B = 90°$ and $BD \perp AC$. If $AD = 4$ cm and $CD = 9$ cm, find the length of altitude BD.

7. **[3 Marks]** In the given figure, O is a point inside ΔABC. Points P, Q, R are taken on OA, OB, OC such that $OP = 2$ cm, $OQ = 3$ cm, $OR = 4$ cm and $PQ \parallel AB, QR \parallel BC$. Prove that $PR \parallel AC$ and $\Delta PQR \sim \Delta ABC$.



SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)

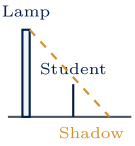


8. **[3 Marks]** In the given figure, $AB \parallel CD \parallel EF$. If $AB = 6$ cm, $CD = x$ cm, $EF = 10$ cm, $BD = 4$ cm, and $DE = y$ cm, calculate the values of x and y.

9. **[3 Marks]** A person 1.65 m tall stands 3.3 m away from a lamppost and observes his shadow to be 1.1 m long on the ground. Find the height of the lamppost.

SECTION C: CASE STUDY BASED QUESTION (4 Marks)

10. **Case Study 06: Night-Time Streetlight & Moving Pedestrian Kinematics**



A municipal street light is mounted at the top of a pole 3.6 m above the pavement. A student of height 90 cm (0.9 m) walks away from the base of the lamppost along a straight sidewalk at a steady speed of 1.2 m/s.

(i) **[1 Mark]** How far from the base of the lamppost is the student after 4 seconds?

(ii) **[1 Mark]** Find the length of the student's shadow after 4 seconds.

(iii) **[2 Marks]** At what rate is the tip of his shadow moving away from the base of the lamppost?

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